15} Immediate Care of The Newborn

Goal:

At the end of this lecture, each student should acquire complete knowledge, skills and positive attitude toward immediate care of newborn and maintain health and save life of newborn.

Objectives:

- **At the end of this lecture, the student will be able to:**
  - Define of immediate care of newborn.
  - Define purpose of immediate care of newborn.
  - Define Apgare score immediate care of newborn.
  - Discuss items of the Apgare score and how to illustrate it.
  - Know how to make physical examination and assessment of newborn.

Outlines:

- Definition of immediate care of newborn.
- Purpose of immediate care of newborn.
- Equipment of immediate care newborn
- Definition of Apgare score.
- Component of Apgare score.
- Physical examination of newborn.
- Reflexes of newborn.
- Procedure of immediate care of newborn
- Reference.

Definition of immediate care of the newborn:

It is care given to the newborn immediately after delivery to keep himhealthy.

Purpose of immediate care of the newborn:

- ☞ To keep the newborn in good health.
- ☞ To assess baby condition.
- ☞ To detected any abnormalities.
- ☞ To avoid asphyxia to the baby.
- ☞ To make card care.
- ☞ To establish and maintain respiratory function.
- ☞ To provide warmth and prevent hypothermia.
- ☞ To ensure safety from injury and infection.
- ☞ To identify actual and potential problems that might require immediate action.

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Equipment:

- ✓ Radiant warmer (servo), Suction apparatus or thin-rubber catheter, endo tracheal tubes.
- ✓ Scissor, alcohol swap, cord clamp, Normal saline, Eye drop, Baby clothes.
- ✓ Alcohol, Paper bag, 6-Stethoscope.
- ✓ 2 towels sterile, Identification sheet.

✓ Vitamin k.

1- Remember the ABC of resuscitation.

A: Air way.

B: Breathing.

C: Circulation.

- Position the infant on the back or side with the head slightly extended inthe neutral position.
- Clear the air way and gently sucks the mucus from the oro-pharynx with a suction catheter or mucus extractor.
- Dry the infant thoroughly with preheated towel to prevent heat loss.
- Initiation of breathing may be stimulated by both drying, and suctioning. However, if the infant does not breath immediately slapping the soles of the feet or rubbing the infant back with a warm blanketing effective measure to stimulate rhythmic breathing.
- If there is no response to tactile stimulation within 10 to 15 seconds, positive pressure ventilation with 100% oxygen, chest compression endotracheal intubation may be indicated.

- Assess the condition of the infant very quickly, using the Apgar score.

- Record the Apgar score.
  - **Suction is stopped when:**
    - Equal bilateral breath sound.
    - Heart rate in normal range.
    - Skin colure is pink.

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Apgar score:

Definition:

It is time schedule measuring the general baby condition to keep baby isgood health.

This is measured after:

1 minute: to assess the general condition of the newborn.

5 minute: to assess the prognosis of the baby.

Sign	0	1	2
Heart rate	Absent	Slow below 100	Fast above 100
Respiration	Absent	Weak crying	Good crying
Muscle tone	Limp of flaccid	Some Flexion	Active flexion
Reflex irritability	No response	Grimace	Good response
Color	Blue and pale	Body pink and extremities blue	Completely pink

*** Total score = 10** *** Normal= 7-10**

*** Mild asphyxia= 4-6** *** Sever asphyxia= 0-3**

Means of identification:

Mother should be notified of the sex of her baby immediately after delivery.

Wrist hand – containing

- Mother name.
- Infant name.
- Infant sex
- Foot print.
- Hospital number.
- Date and time birth.
- Race, eight & length.

Male circumcision:

This should not be performed immediately after delivery but rather a day or two later, after the infant has been demonstrated to be healthy.

The baby's bath:

- At birth baby's skin is covered with greasy vernix casosa which providesa natural barrier against infection.
- But the excess of tit in the neck folds andgroin may cause irritation off the skin.
- Bathing the baby immediately after delivery is not recommended; only wipe baby's skin with sterile moistened swab before clothing him.
- One competes bath given by the mother under instruction, on the second or third day before discharge is all that is required.

Physical examination:

Must be through to assess the newborn, assess any congenital anomalies, birth trauma and specific neonatal problems.

General appearance:

Measurements

(Weight): - ranges from 2.700 -3.500 gm, and average of 3 gm.

- **First 3 day:** - lose 5-10% from birth weight.
- **7-10 days**: - region birth weight.
- **Six months**: - double his weight

(Length): - average length of male and female (47.5-53.5cm), the newborn will grow approximately and inch a month for the six months.

(Head circumference):- 33-35 cm.

Vital signs (especially Blood pressure)

Skin

During the examination, the condition of the skin should be observed, the color should be noted.

- Color: normal is pink, observation for cyanosis, pallor or

jaundice.

- Presence of edema and skin turgor.
- Lanugo hair, found hair on the shoulder neck.
- Observe for rashes or spots and his color.

Vernix caseosa:

- Whitish greasy material.
- Covering the skin, disappear.
- Dry after several hours.

Head

- **Normal Head**: round, symmetrical and moves easily from left to right andup and down.

- **Moulding**: the amount of overring or spreading of the suture is noted.
- **Anterior Fontanelle**: it is diamond shaped closed at 12-18 months bulging
- anterior fontanelle may indicate intracranial hemorrhage or meningitis.
- **Posterior Fontanelle**: triangle-shaped closed t birth or within first 2 months.
- Delaying closure occur in congenital hypothyroidism.
- **Cephalohematoma:** resulting from the trauma during labor.
- **Caput succedaneum**: is edema of the scalp due to

the pressure causedduring labor and delivery.

Eyes:

- Conjunctival or sclera hemorrhages are common and usually are ofno clinical significance.
- Edematous for few days of live, resulting from delivery.

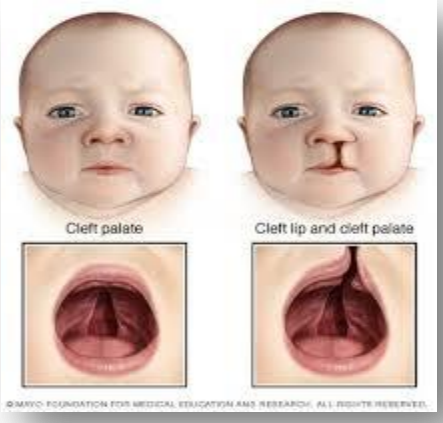
Mouse and Nose

Nose

- The shape of the nose and width of the bridge should be observed, this should be greater than 2-5 cm in the term baby.
- Obstructions of the nasal passage result in respiratory distress.

Mouth:

- ☞ The mouth should be examined by inspection and palpation.
- ☞ Mouth draws to one side resulting in facial nerve paralysis.



- ☞ Prematurely erupted teeth may be present in the lower.
- ☞ Cleft palate.
- ☞ A small mouth may be due to

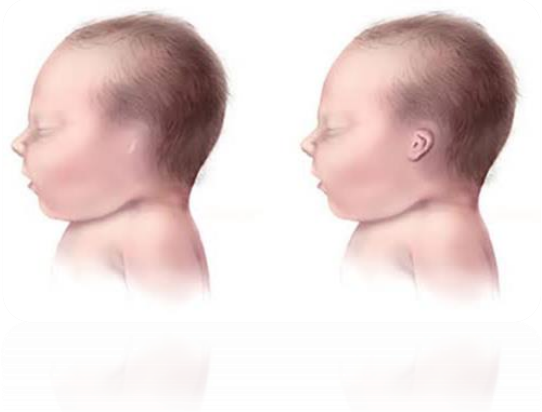
micro-genitalia.

Ears:

- Examine the ear to ensure two are present and they are fully formed, in the correct position, the ears of the term baby should contain enough cartilage.
- Observe for lesion or nodules on external ear.
- Infant reacts to loud voice.
- Inner canal is present, no discharge.
- Malformed or low set ears may be associated with congenitalanomalies.

Neck:

- Babies have short necks, which should be examined for symmetry, move the fingers around the neck to detect the presence of any swelling.



- Observe neck for masses or goiter.
- Abnormally short neck.
- Inability to flex neck indicate to meningitis.

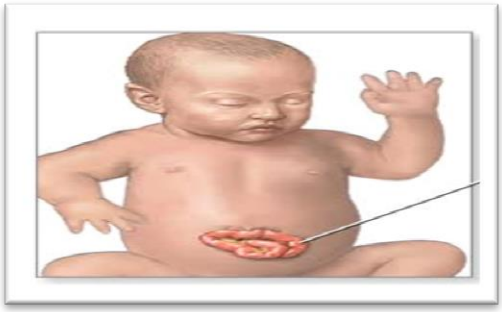
Thorax:

- Observe size symmetry and shape.
- Observe breath sounds and respiratory

rate.

- Observe heart sound, rhythm and rate.
- Check breasts for engorgement.

Abdomen:



- Observe shape.
- Observe umbilicus and cord vessels.
- Check femoral pulse.

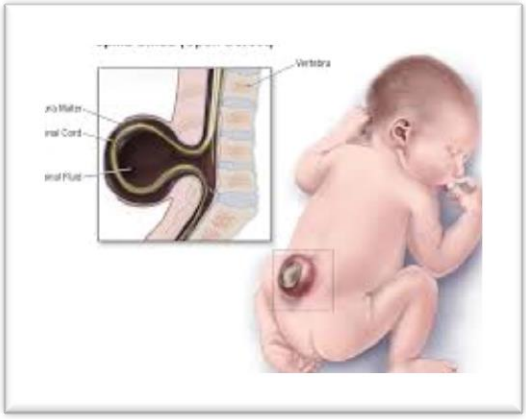
Genitalia:

- Observe for appropriateness of sex.
- Observe female labia and discharge.
- Observe male urethral opening, scrotum and testes.
- Observe urine for amount and color.

Back:

- Observe spine for symmetry of vertebrae.
- Check anus potency and meconium.

Extremities:



- Observe for symmetry, movement and abnormalities.
- Observe digits of hands and feet for number and feet for number andwebbing.

- Observe hips for range of motion.
- Observe feet for abnormalities.

Breast feeding:

- The infant should be given to the mother for breast feeding. Early breast feeding is recommended to prevent hypoglycemia, promote mother infant bonding and stimulate secretion.

Reflexes Examination

New natal reflexes are found during the first few months of postnatal life absence of these reflexes indicate to failure of central nervous system development and maturation.

1. **Rooting reflex:-**
   - Turn in direction of stimulus to cheek or mouth.
   - Disappears by 7 months of age.
2. **Sucking reflex:**
   - Opens mouth and begins to suck when put finger or nipple.
   - Disappears by 12 months of age.
3. **Palmar grasp reflex:**
   - Fingers grasp adult finger when palm is stimulated and hold momentarily.
   - Disappears by 3-4 months of age.
4. **Planter grasp reflex:**
   - Toes curl down ward when sole of foot is stimulated.
   - Disappears by 8 months age.

5. Tonic neck reflex (fencer position):

- When head is turned to one side extremities on same side extend and on opposite side flex. Disappears at 3-4 months of age.

Immediate Care of Newborn Guideline

Action	First Trial		Second Trial	
	Done	Note Done	Done	Note Done
Getting Ready				
1. Prepare the equipment: ☞ Radiant warmer and warm receiving blankets ☞ suction apparatus with sterile catheter ☞ Oxygen mask with three different sizes. ☞ Laryngoscope with different sizes blades OR airway. ☞ Endo- tracheal tubes with different sizes and satellite. ☞ Sterile gloves ☞ Cord care tray. - Sterile scissor. - Sterile clamp. - Dry cotton and gauze - Cotton with alcohol.				

- Paper bag. ☞ Vitamin K. ☞ Neonatal health card. ☞ Baby scale, measuring tape 2. Prepare the servo (heater) as the following: ☞ Connect the plug into the electricity source. ☞ Get the warmer and the light bottoms on. ☞ Cover the mattress with new linen. 3. Prepare the suction apparatus as the following: ☞ Ensure the apparatus function. ☞ Remove any remaining secretion from the suction jar. ☞ Connect a new suction catheter without removing its cover				
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without removing its cover.				
5. Prepare the oxygen apparatus as the following: ☞ Ensure its functioning. ☞ Ensure that the level of distilled water is between the minimum and maximum level.				
Immediate Care of The Newborn Steps				
6. Wash hands.				
7. Wear the sterile gloves.				
8. As soon as the infant is born receive him in a clean blanket or towel to maintain warmth and to prevent heat loss and hypothermia.				
9. Transfer the baby to the heater.				
10. Provide warmth by keeping the baby dry (start to dry the head, face and the rest of the baby by dry towel) to minimize loss of infant's body heat.				
11. Be sure that the baby is dry from any amniotic fluid and avoid air drafts in the room.				

12. Use the small rubber suction catheter which is attached to a mechanical suction to remove aspiration fluids that may be obstructing the air way. 13. Suction is performed through oropharynx. 14. Maintain a modified Trendelenburg position to facilitate drainage of secretions 15. Suction the newborn's mouth then nose.				
16. Give the newborn oxygen (6-8L) through oxygen mask with raising the chain of the newborn in between suctioning and as needed.				
17. Assess the newborn's condition immediately after birth (at 1 minute) through Apgar score (heart rate, respiratory effort, muscle tone, reflexes and color) and is repeated at 5th minutes Apgar score is an index that provides simple accurate and safe means of quickly appraising the baby's condition.				

☞ Identification 18. Provide identification of the infant by applying ID band to the arm or leg, which includes mother's name, hospital number, newborn sex, time and date of birth 19. Take the newborn's foot prints according to the hospital policy				
Cord cutting and care as the following: 20. Clamp the cord after 2 fingers "4 cm" from its insertion in the abdomen then cut it after one finger "2 cm" above the clamp. 21. Clean the umbilical cord and its base with cotton ball soaked with alcohol after putting the sterile dressing around its base. 22. Ensure that the cord has 2 small arteries and one big vein				
23. Evaluate the newborn condition again at fifth minute using Apgar scoring system (HR measured in full minute).				
24. Eye care using antibacterial agent. i-e- erythromycin to protect the eyes				

against infections. Wipe each eye from inner side to outer side in one direction using sterile cotton balls. Apply 1-2 eye drops of antibacterial drops				
25. A single dose of vitamin K is administered intramuscularly in the vastus lateralis muscle as a preventive measure against neonatal hemorrhagic disease. 26. Measure head, chest circumferences and length using a measuring tape. • **Head circumferences:** across eye brows of the newborn. • **Chest circumferences**: across breast nipples of the newborn. • **Length:** measured by using hard object behind the head of the newborn to insure correct measurement. **Weight**: use baby scale for weighing the baby after check the accuracy through balancing zero prior to obtaining the weight.				

27. Assess the skin for scaling, undue wrinkling or birthmarks, lanugo, vernix, texture or rash, color to indicate the maturity of the baby and exclude any jaundice or abnormalities.				
28. The abdomen is palpated for early detection of any mass, enlargement, and congenital hernia, swollen organ (liver, spleen, and kidney).				
29. Examine the appropriateness of sex: ☞ The genitalia for male urethral opening, scrotum and tests. ☞ Female labia and discharge.				
30. Check the extremities, shoulders movements, abnormalities of hands, feet, and extra fingers.				
31. Provide suitable dress.				
32. Take the newborn away from the heater.				
Post Procedure Tasks				
33. Clean and remove all equipment				
34. Wash hands				
35. Record your findings into the proper				

card				
36. Assess the skin for scaling, undue wrinkling or birthmarks, lanugo, vernix, texture or rash, color to indicate the maturity of the baby and exclude any jaundice or abnormalities.				

15} Immediate Care of The Newborn Checklist

Action	First Trial		Second Trial	
	Done	Note Done	Done	Note Done
Getting Ready				
1. Prepare the equipment needed.				
2. Wash hands.				
3. Wear sterile gloves.				
Immediate Care Of The Newborn Steps				
4. Dry & warmth the infant thoroughly.				
5. Suction the newborn in a Trendelenburg position.				
6. Give the newborn oxygen through oxygen mask.				
7. Evaluate the newborn's condition using Apgar scoring system at 1st minute.				
8. Identification (apply ID band, foot prints).				
9. Perform cord cutting, care and assessment.				
10. Evaluate the newborn's condition using Apgar scoring system at 5th minutes.				
11. Eye care				
12. Administration of vitamin K.				
13. Perform measurements (weigh, head, chest circumference and length).				
14Check of neonatal neurological reflexes.				
15. Routine care (general physical examination, and provide suitable dress).				
Post Procedure Tasks				
1. Remove equipment.				
2. Wash hands.				
3. Documentation.				

Score		
Date:		

Student signature: Instructor signature: